Problem Statement and Goals
Keystone

Team 13

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Table 1: Revision History

Date	Developer(s)	Description of Changes
Sept 25th 2026	Ayush Patel	Added Problem Statement and Custom Documents
Sept 26th 2026	Waqar Ul-Hassan	Added Goals and Stretch Goals
Sept 26th 2026	Waqar Ul-Hassan	Added reflection information
Sept 27th 2026	Waqar Ul-Hassan	Updated Goals and Stretch Goals based on feedback
Sept 27th 2026	Ayush Patel	Added reflection
Sept 28th 2026	Ahmed Rashrash	Added reflection information
Sept 28th 2026	Andrew Habib	Added reflection information
Sept 28th 2026	Ayush Patel	Minor Formatting and Proofreading changes
Oct 25th 2026	Ayush Patel	Updated Problem Statement and Goals

1 Problem Statement

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1.1 Problem

Capstone team and project selection at McMaster, across the Software Engineering (SE), Computer Science (CS), and Mechatronics (TRON) capstone courses, is currently fragmented across multiple disconnected tools, including Teams,

Avenue to Learn, GitLab, MS Forms, and email. This fragmentation creates administrative overhead, frustration, and a lack of transparency for everyone involved. Beyond the fragmentation itself, the information within these tools is often unreliable.

Project descriptions vary widely in accuracy and level of detail, making it difficult for students to compare opportunities and understand what a project actually involves. Due to this lack of clarity, students can lose valuable time early in the process, which can be critical given the limited duration of an eight-month project. Supervisors, in turn, may become reluctant to offer projects in future years if expectations are consistently miscommunicated, reducing the range of opportunities available to future cohorts.

Furthermore, students have limited visibility into which projects are still available, and there is no unified way to track a project's status from initial submission through final team assignment.

Beyond finding a project, students also struggle to find teammates who share similar skills, work styles, and interest in the same projects. Team formation currently happens informally, through personal networks or chance, rather than through any structured process.

Supervisors face related difficulties. They receive little guidance on what a capstone project should look like before proposing one, project ideas proposed by students are not visible to them, and they know little about the students and teams who contact them. There is also no shared record of which teams and supervisors have agreed to work together.

Instructors, in turn, have no view across the capstone courses, so they cannot tell which proposed projects went unselected and follow up with those supervisors. Reviewing team submissions involves giving the same feedback to many teams, and team and project information often changes after approval without the official records being updated.

Together, this lack of transparency, unreliable information, and absence of structured team formation lead to wasted time, mismatched teams, and added administrative overhead for course instructors and TAs.

1.2 Inputs and Outputs

Inputs:

- Project proposals and descriptions submitted by supervisors and students
- Student profiles
- Team information, including team membership
- Student expressions of interests and applications to projects

- Course configuration and information (e.g., team formation rules, schedules, guidelines)
- Instructor decisions and approval outcomes recorded by instructors or TAs (approvals, requests for changes, status changes)
- Changes to previously approved team and project information

Outputs:

- A searchable and filterable catalog of available projects
- Project status and availability information
- Confirmed project–team assignments
- Records of agreement between supervisors and teams
- Visibility into other students seeking teammates
- Notifications of events requiring a user's attention
- A list of projects that were not selected
- A validated record of team and project information

1.3 Stakeholders

- **Capstone Students** — The primary users of the platform. They browse and filter available projects, express interest or apply, and form teams. Their main need is clear, accurate, and up-to-date information so they can make informed decisions and avoid wasted effort pursuing unavailable or misrepresented projects.
- **Capstone Project Supervisors** — Professors or industry sponsors who submit and manage project proposals. Their main need is a simple, standardized way to describe their project, specify how many teams they are willing to take on, and manage student interest without relying on ad hoc email communication.
- **Capstone Course Instructors** — Oversee the capstone process for their course. Need visibility into which projects and teams exist across the capstone courses and their status from proposal through approval, including later changes to team and project information, and administrative control to resolve issues.
- **Capstone Course TA's** — Support the instructors in administering the process. Need efficient tools to help manage day-to-day tasks such as verifying project submissions, tracking applications, and assisting students and supervisors.

1.4 Environment

Software — The application is accessible through any modern web browser. An internet connection is required to use the platform.

Hardware — The application is designed to run on standard desktop and laptop computers, as well as tablets and smartphones. It does not require any specialized hardware.

2 Goals

The primary goal of the platform is to provide a centralized and structured system for managing the capstone project process. The platform will improve the experience for students, supervisors, instructors and TA's by making project information easier to access, manage, and coordinate.

2.1 Core Goals

1. **Centralized and Consistent Project Information:** Provide a single platform where capstone projects, whether proposed by supervisors or by students, can be submitted, viewed, managed, and tracked from proposal through team assignment and any later changes. Projects are described in a consistent submission format in which a defined set of required information, such as objectives, requirements, required skills, and expected deliverables, must be provided.
2. **Project Discovery:** Allow students to search and filter available projects based on factors such as skills, technologies, project type, industry, supervisor, and availability.
3. **Student Profiles and Team Formation:** Allow students to create profiles containing relevant information such as skills, interests, experience, and project preferences, and to indicate their intent to form or join a team. Provide visibility into other students seeking teammates for the same or similar projects, and allow supervisors to view the profiles of interested students and teams.
4. **Project Interest and Application Management:** Provide students with the ability to express interest in projects and allow supervisors as well as sponsors to review and manage interested students.
5. **Course Information and Guidance:** Provide supervisors and students with timely information about the capstone courses, such as expectations, timelines, and examples of past projects, so that projects can be proposed and selected with a clear understanding of what is expected.
6. **Instructor Oversight and Administration:** Provide instructors and TAs with tools to manage users, projects, project availability, and project assignments, with visibility into projects, teams, and their status across

capstone courses, including projects that were not selected. Support instructors in approving projects and teams and in keeping team and project records accurate.

2.2 Stretch Goals

The following features may be implemented if sufficient development time and resources are available.

1. **Team Compatibility Matching:** Recommend potential teammates with respect to their skills, interests, and project preferences.
2. **Intelligent Project Recommendations:** Create a recommendation system that suggests suitable projects to students based on the overlap between the student's skills, interests, and preferences and a project's requirements, such that suggestions are demonstrably more relevant than random ones.
3. **Analytics Dashboard:** Provide instructors and TA's with statistics regarding project demand, student engagement, applications, and project availability.

3 Custom Documents (CDs)

- **User/Stakeholder/Client Interview Report** – Documents the interviews conducted with users, stakeholders, and clients to gather requirements for the application.
- **Usability Testing Report** – Documents the usability testing of the application, including the goals of the testing, the specific tasks we ask users to complete, the specific questions we ask users, and the results

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

Waqar Ul-Hassan:

1. What went well while writing this deliverable?

   The team has been awesome in picking up work and I think that's been the best part of it so far. We haven't had any issues with assigning work and infact most of us have been trying to get more work to make sure that we are all keeping up. Furthermore, being able to get to know my team members are bit better as I didn't know two of them before this project has been nice as well. We also had to drop our original project we had committed to and I think our team recovered from that pretty well, I was a bit worried initially but we got everything together quite quickly!

2. What pain points did you experience during this deliverable, and how did you resolve them?

   I had a couple pain points through this process, the first was of course the fact that our first project ended up being somewhat complete already which was something that caught us off-guard. I Feel as though we did a great job in resolving this by communicating with our supervising professor as well as Dr. Smith and then ensuring we had a backup project incase we wanted to switch. Another Pain point I had was when deciding some of the metrics for how to track all the work being done equally. The solution to this was just holding discussions, we discussed metrics like commits, number of issues, and lines of code. Overall, we ended up deciding on the number assigned issues as the metric because we felt we could best distribute that one.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

   We initially had some more particular problem solutions for our project

which we faced personally in finding a supervisor, projects, and navigating through all the information in general. After the discussion with our TA Chris I realized that maybe we needed one or two more smaller things that we should aim to reach as part of our main goal. With this in mind we had discussions with each other and decided on what our goals and stretch goals are now. I also realize as we meet with stakeholders these requirements could change and our TA had said that in the case that they do, we can discuss it with him to update our goals to ensure everything makes sense.

Ayush Patel:

1. What went well while writing this deliverable?

   It was really easy for us to come up with problems and goals because this is something we've actually experienced as a team. Our initial project fell apart because the supervisor's expectations didn't match ours, and on top of that we had a hard time figuring out which projects were still open. That first-hand experience made it really easy to write the problem, goals, and stakeholder sections since we weren't just guessing at pain points, we lived them.

   Given the nature of our project, it was also pretty simple to figure out which custom documents we needed. We'll definitely need to interview a bunch of people (students, supervisors, instructors) to nail down requirements, and we'll also need usability testing to make sure we actually built something usable by all three groups.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   The main pain point was figuring out how to keep our goals abstract enough that they didn't just turn into requirements in disguise. We kept catching ourselves writing things like "the system shall do X," which is way too detailed for a goal. We resolved this by going back and forth as a team, asking "is this describing a capability, or is this describing how it works," and rewording anything that leaned too far into implementation.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

   In our initial discussion with our TA, it came up that our scope might be a bit too small for a capstone project. But if we added a full team-matchmaking algorithm as core scope, that would make things too big. So we split the difference: basic team formation (letting students indicate they want a team and see who else is looking) became a core goal, and the smarter, algorithm-driven matchmaking became a stretch goal instead.

Ahmed Rashrash:

1. What went well while writing this deliverable?

   Tasks were split fairly and all work that needed completing was accounted for. All group members participated when reviewing work and suggestions were productive.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   The biggest issue during this deliverable was encountered before even starting it. After sifting through many externally-offered projects, we thought we landed on a great project which gave us full freedom in designing and implementing it. However, we learned that major components were already designed, implemented, and tested, leaving little work to be done. We were able to resolve this issue by meeting as a group and with the supervisor, weighing options, taking all opinions and facts into consideration, and eventually reaching a consensus. We decided to use this exact problem as our capstone project. Our team's ability to quickly pivot allowed us to naviagate a big roadblock with ease.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

   Our team drew on our knowledge, skills, and past experiences, and used that to determine how fast specific key features and systems could be designed, implemented, and tested, as well as their expected quality. We also factored in whether certain sub-problems have been solved before to decide whether certain problems needed additional considerations. For example, recommendation systems have been developed many times in the industry but are still an active area of research and were classified as a stretch goal. We also actively communicated our ideas with professors and TAs, and as a result expanded our core goals to ensure they provided enough substance.

Andrew Habib:

1. What went well while writing this deliverable?

   During this deliverable, our team was able to showcase their resilience despite the challenges we faced while selecting a project. There was great communication amongst each other to ensure that feedback was given and we were all on the same page regarding our goals for the project. Everyone was active and participating in the brainstorming processes for the project. The goals were well formed and clear when we discussed them as a group especially when we transitioned to our own idea. Once our goals were established, the inputs and outputs, stakeholders, and environment were obvious. Overall, I think our group is coming together well and I am excited to take on this project with them.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   The biggest pain point that we faced came in the selection process of our project (hence, prompting the idea of our capstone project), where the initial idea fell through due to a misunderstanding on the level of completion of that project. This prompted some disagreements among our group as we all tried to evaluate the best course of action (whether to continue or not). The way we overcame this was through a thorough analysis of the pros and cons. Everyone's opinion was valued and we ultimately decided that we needed a new project. Our group was able to resolve this issue very quickly by resubmitting a new idea on the same day. In terms of the deliverable, some goals were challenging to determine this early but group discussions allowed us to come together to figure out the main ideas we wanted to focus on for now.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

   The way we kept things manageable and complex enough was through communication with idea evaluation using prior experience as well as communication with our TA. We realized that our idea could be either very simple (if we were to simply implement a project posting web page) or too complex (if we were to implement an advanced recommendation system). Communication with our TA allowed us to add a couple necessary features on top of our basic goals to form a complete project while keeping complex features as stretch goals.